The Acceptor-Recognition Circuit

Extension to the SpliceBERT Donor-Circuit Proposal — work by Sribhanu Gundu

Repository	`gfm-circuit-discovery/acceptor-interpretability/`
Branch	`sri-algoverse-acceptor-interpretability`
Author	GunduSriBhanu <sribhanu5june@gmail.com>
Model	SpliceBERT.1024nt — 6 layers × 16 heads, hidden size 512, post-LayerNorm BERT-MLM, absolute learned position embeddings
Data	GUE `splice_reconstructed` (GENCODE/DeepSplice-derived), real human 3′/5′ splice-site windows, n=200 acceptor samples, fixed seed
Report date	2026-08-10

1. Motivation and question

The SpliceBERT donor-circuit proposal established how the model reconstructs a masked **donor** site (5′ `GT`) using information from its paired 3′ **acceptor** site — but found that the branch point (BP) and polypyrimidine tract (PPT) region contribute ~0 to that donor-side payload. This project asks the reciprocal question left open by that result: **where and how does the model use the BP/PPT signals to recognize the acceptor itself?** The acceptor representation this circuit builds is the payload the donor circuit currently takes as given — this work slots upstream of the donor circuit's central head (L3H1) without disturbing that result.

Metric: `m_acc` — masked-AG logit margin. The acceptor's `AG` dinucleotide is masked; the model's confidence for the true base at each position is scored against the other three nucleotides and summed across both positions. This mirrors the donor circuit's own `m(x)` metric, making results directly comparable in style.

2. Method summary

- **Content-destroying corruption** — each corrupted base redrawn from a balanced ACGU distribution via a Markov process that provably cannot introduce an `AG` substring anywhere in the region or at either seam with flanking real sequence.
- **Composition-preserving shuffle-in-place** — same bases, reordered — isolates whether an effect is driven by base *order/identity* or bulk *composition*.
- **Random-AG floor control** — destroys the entire window, forces a bare `AG` back in at the true position (zero genuine context).
- **Far-intronic control** — content-destroying corruption deep in the intron, away from any splice signal; the honest noise floor.
- **Downstream decoy-AG control** — tests whether the model prefers the correctly positioned site over an incidental downstream `AG`.
- **Per-sequence branch-point localization** — a hand-set `yUnAy`-consensus PWM scan over the canonical 18–45nt upstream window (self-contained stand-in for branchpointer/LaBranchoR).

- **Per-head attention-mass readout** — at the masked-AG query position, into the BP/PPT regions, under clean vs. corrupted vs. shuffled conditions.
- **Causal intervention ladder** — mean-ablation (necessity), restoration/mediation patching, and Q-patch/V-patch path-patching split (routing vs. content; Wang et al. 2022, Heimersheim & Nanda 2024, Goldowsky-Dill et al. 2023), via PyTorch forward hooks on `SpliceBertSelfAttention`.
- All effects reported with 5000-resample bootstrap 95% CIs.

3. Results by phase

Phase 0 — Pilot (real data, n=200 acceptor / n=200 non-splice)

Comparison	Δ (mean)	95% CI	Acceptor-specific?
BP disruption alone	+0.212	[+0.061, +0.360]	yes — control −0.105 [−0.201, −0.027]
PPT disruption alone	+1.245	[+0.875, +1.624]	yes — control +0.046 [−0.171, +0.267]
BP+PPT both disrupted	+1.386	[+0.999, +1.775]	yes — control −0.145 [−0.438, +0.125]
Context required (clean − random-AG)	+1.319	[+0.919, +1.739]	—
AG-scanning (true − downstream decoy)	+0.916	[+0.478, +1.362]	—

Context is required (largest effect); AG-scanning is supported; BP disruption lowers `m_acc` and is acceptor-specific. **Unexpected:** PPT disruption alone has a large, clearly acceptor-specific effect — not the "does ~nothing" result a synthetic pilot had reported. BP+PPT together (+1.386) is close to additive (1.245+0.212=1.457), not the AND-gate signature originally hypothesized.

Phase 1 — Control fix: is it real, and is the tract effect order or composition?

Control	Δ	95% CI	Reads zero?
Far-intronic (content-destroying)	+0.046	[+0.011, +0.085]	technically no, but 5–27× smaller than every real effect — the noise floor
PPT shuffle-in-place (composition preserved)	**+1.207**	[+0.897, +1.530]	**no — nearly as large as full corruption (+1.245)**
BP shuffle-in-place (composition preserved)	+0.168	[+0.041, +0.305]	no, but small

> **Key finding:** PPT shuffle-in-place was expected to read ~zero (same bases, same position) and instead came back at +1.207 — almost the full content-destroying effect. This overturns a "bulk pyrimidine density" reading: whatever SpliceBERT reads in that window is **order/identity-sensitive**, not aggregate composition. The branch-point effect (+0.212) sits right at the edge of the corrected floor's CI — the better control didn't clear it, it just replaced "looks real" with "unresolved."

Phase 2 — Absolute floor: gate vs. artifact

Condition	Absolute m_acc	95% CI	vs. floor
Clean	+0.655	[+0.254, +1.072]	—
BP disrupted	+0.444	[+0.037, +0.855]	**clearly above floor**
PPT disrupted	−0.590	[−0.842, −0.331]	**at floor**
BP+PPT both disrupted	−0.730	[−0.935, −0.515]	**at floor**
Random-AG floor	−0.664	[−0.728, −0.595]	—

PPT disruption alone already crashes `m_acc` down to the no-context floor. BP-only stays well above the floor. "Both disrupted" collapsing to the floor is fully explained by PPT alone — not a BP×PPT interaction (AND-gate). Simpler picture: **the tract is close to sufficient on its own; the branch point contributes a smaller, dispensable increment.**

Phase 3 — Real coordinates: relocate BP/tract per sequence

Effect	Fixed-offset (Phase 0)	Real-coordinate (Phase 3)
BP-only disruption	+0.212 [+0.061, +0.360]	**+0.090 [-0.007, +0.186] — CI now includes zero**
PPT-only disruption	+1.245 [+0.875, +1.624]	+1.433 [+1.076, +1.802] — larger, not smaller
BP+PPT both disrupted	+1.386 [+0.999, +1.775]	+1.498 [+1.083, +1.914]

Per-sequence BP localization (mean tract length 27nt vs. fixed 15nt; only 2/200 sequences had no candidate) did **not** rescue the branch-point effect — if anything it weakens and loses significance, while the tract effect strengthens. Rules out "the BP window was just mislocated" as the explanation for the earlier small effect.

Phase 4 — Head content-control, retargeted to tract-readers

Tested whether the six candidate heads carried over from the donor-circuit proposal (`L0H2, L1H14, L2H7, L2H0, L3H10, L1H1`) are genuine content readers, or attend there purely by positional locality.

Original 6 candidates — all fail

Head	PPT clean mass	PPT verdict	BP clean mass	BP verdict
L0H2	1.000	positional-locality artifact	~0	no attention
L0H14	1.000	positional-locality artifact	~0	no attention
L1H14	0.999	positional-locality artifact	~0	no attention
L2H7	0.993	positional-locality artifact	~0	no attention
L2H0	0.968	positional-locality artifact	0.009	no attention
L1H1	0.903	positional-locality artifact	0.028	no attention
L3H10	0.815	positional-locality artifact	0.041	no attention

All six spend 80–100% of their attention budget on the 15nt immediately upstream of the masked-AG query — but this mass is statistically unchanged (sometimes higher) when content is destroyed. Textbook previous-token/local-window head signature, not a tract reader.

15 newly discovered candidates (10 top BP-attenders + 5 top PPT-attenders, found by ranking clean attention mass) → 5 pass PPT content-control

Head	Clean PPT mass	Δ(clean−corrupt)	Δ(clean−shuffle)	Order-sensitive?
L5H8	0.076	+0.0223 [0.017, 0.027]	+0.0080 [0.005, 0.011]	**yes**
L4H12	0.063	+0.0119 [0.008, 0.016]	+0.0032 [0.0002, 0.006]	**yes**
L3H5	0.081	+0.0087 [0.0045, 0.013]	+0.0023 [-0.001, 0.006]	no (CI incl. 0)
L0H13	0.383	+0.0009 [0.0004, 0.0014]	+0.0003 [-0.0001, 0.0008]	no (CI incl. 0)
L0H14	1.000	+0.00001 [0.0000075, 0.0000142]	+0.00001 [0.0000088, 0.0000145]	yes, negligible (saturated)

L5H8 and L4H12 are the clearest finding: real, order-sensitive attention drops (not just composition), consistent with Phase 1's order/identity-driven tract signal — the first positive evidence in this project of heads that plausibly *read* the tract. Their clean attention mass (6–8% of budget) is far below the saturated positional-lock heads (80–100%), so they are not the dominant attention sink, but the content-sensitivity is real and well-powered.

Branch point: zero of the 21 tested heads pass content-control. Several top BP-attenders (`L2H12, L5H8, L5H14`) even show attention *increasing* under corruption or shuffle — the opposite of a content-reading signature.

Phase 5 — The intervention ladder: necessity, mediation, routing vs. content

Tests whether the content-tracking *attention* found in Phase 4 translates into *causal* effect on `m_acc`. Applied to `L5H8` , `L4H12` (order-sensitive hits), `L3H5` , `L0H13` (ablation only), and negative control `L0H2` (most attention-saturated donor-circuit head, 0/6 that passed content-control).

Step 1 — Mean-ablation (necessity)

Head	Δ m_acc (clean − ablated)	95% CI	Significant?
L5H8	-0.004	[-0.015, +0.007]	no
L4H12	+0.001	[-0.009, +0.012]	no
L3H5	-0.014	[-0.030, +0.0003]	no (edge)
L0H13	+0.003	[-0.005, +0.011]	no
L0H2 (control)	-0.015	[-0.031, +0.0003]	no (edge)
Joint: 4 readers together	-0.015	[-0.041, +0.010]	no
Joint: all 5 together	**-0.030**	**[-0.060, -0.002]**	**yes, barely**

Zero heads individually pass necessity — every CI straddles 0, including the control. Jointly ablating all 5 barely clears significance at −0.030, about **2% of the size of the real PPT-corruption effect (+1.2 to +1.4)**.

Step 2 — Restoration patching (mediation)

Head → recipient	Recovery fraction	95% CI
L5H8 → corrupted	+0.1%	[-0.6%, +0.8%]
L5H8 → shuffled	-0.2%	[-0.7%, +0.3%]
L4H12 → corrupted	+0.3%	[-0.3%, +0.9%]
L4H12 → shuffled	+0.0%	[-0.2%, +0.4%]

All four recovery fractions are indistinguishable from 0%. Patching a single head's true output back in recovers essentially none of the corruption effect.

Step 3 — Q/V split (routing vs. content)

Head	Q-patch (routing) recovery	V-patch (content) recovery
L5H8	-0.1% [-0.4%, +0.2%]	-0.2% [-0.8%, +0.5%]
L4H12	-0.2% [-0.4%, +0.03%]	+0.2% [-0.4%, +0.8%]

Both null too — there's no effect left to decompose once the whole-head patch was already null.

> **Verdict:** single- and small-group causal necessity is a clean null, contradicting a naive reading of Phase 4. These heads' attention genuinely tracks tract content (real, well-powered, order-sensitive), but that correlation doesn't cash out as a large individual causal channel to `m_acc` through this one layer's direct output at the AG position. The tract-reading computation is either (a) genuinely distributed across many more components than these five (a "hydra effect"), or (b) routed indirectly — a head could write tract information to a position other than the AG query, read by a later layer, invisible to a layer-local intervention.

4. Overall mechanistic headline

- Context and AG-scanning are robust and unambiguous — **GO** on the overall direction.
- **The polypyrimidine tract is the dominant, close-to-sufficient driver** of acceptor confidence; the signal is order/identity-driven, not bulk composition.
- **Five heads** (`L5H8, L4H12, L3H5, L0H13, L0H14`) show genuine, content-sensitive attention into the tract — a first positive behavioral hit; `L5H8/L4H12` are order-sensitive. None were on the original candidate list.
- That content-sensitivity **does not translate into individual causal importance** — Phase 5's intervention ladder is a clean null, even jointly across all five heads (~2% of the real effect).
- The branch-point-alone effect has **not survived four independent attempts** to firm it up (matched floor, absolute-vs-floor test, per-sequence relocation, and a full 21-head attention content-control sweep) — treat as unresolved/likely weak, not secondary-but-real.
- The original AND-gate framing is not supported — Phase 2 shows PPT-alone already reaches the floor, sufficient on its own to explain "both disrupted."
- The six donor-circuit candidate heads are a **dead end** for this problem — all six are saturated positional-locality heads; none read content at either region.

5. Known approximations / caveats

- The original GUE `label==2` non-splice control (Phase 0) is superseded by Phase 1's far-intronic control — it wasn't reliably intronic, likely explaining its small but significant −0.105 reading instead of zero.
- The branch-point locator is a hand-set `yUnAy`-consensus PWM scan, not branchpointer or LaBranchoR — Phase 3's null result is a claim about *this locator's* candidate positions, not proof the branch point is biologically irrelevant.
- The decoy-AG check only uses samples where a downstream AG exists within the search window.
- All phases run on the same n=200 fixed-seed sample for comparability — not yet checked for stability at larger n; some CIs are wide enough that a larger sample could move marginal calls (e.g. real-coordinate BP-only: [-0.007, +0.186]).

6. Next steps (from the project's own roadmap)

1. **Broaden the causal search** — path-patch backward from `m_acc` itself (not from Phase-4's attention ranking), letting the model nominate which components matter (à la Wang et al.'s IOI methodology) rather than filtering by attention first.
2. **Include MLP outputs and other token positions** — a head could write tract information somewhere other than the AG query, for a later layer to read.
3. **A true whole-circuit scrub** (ablate the full attention layer or several layers jointly) as an upper bound on how much any causal story could explain, to calibrate how far off the 2% figure actually is.
4. **Shared-transport question** — does whatever mediates the tract effect (once found) reuse the donor spine's transport heads (L4H0, L4H2, L5H11) and a `U_info`-like subspace, or run on its own? Deferred until a mediating component is found.
5. **Real branch-point annotations** (branchpointer / LaBranchoR / Leman et al.) in place of the PWM stand-in, plus gene-disjoint held-out splits — the one thing that could still revive a standalone branch-point finding.

7. One-line status

> Real data is a **GO** on the overall direction (context and AG-scanning are robust); the mechanistic headline has moved from branch point to polypyrimidine tract; Phase 4 found heads whose *attention* genuinely tracks tract content, but Phase 5's intervention ladder found that content-sensitivity doesn't cash out as individual or small-group causal necessity — the real circuit is more distributed or more indirect than five heads, and finding it is now the open problem; the branch-point-alone effect has not survived four independent checks and should be treated as unresolved.

Work by Sribhanu Gundu (GunduSriBhanu) — `gfm-circuit-discovery/acceptor-interpretability/`, branch `sri-algoverse-acceptor-interpretability`. See the project's own `README.md` for full layout, setup instructions, and reproducibility details.